Classical Mechanics 1: Mid-semester Examination

Marks: 100

Name: _____ **Roll no.:** _____

1. Suppose that a projectile is subject to a linear resistive force $\mathbf{F}_d = -b\mathbf{v}$ and is thrown vertically downward with speed v_0, where $v_0 > v_{\text{ter}}$. Take the positive y-direction to be vertically upward. Describe and explain how the velocity varies with time, and make a plot of v_y against t for the case $v_0 = 2v_{\text{ter}}$. [15]

2. Show that if the net force on a one-dimensional particle depends only on position, $F = F(x)$, then Newton's second law can be integrated to give v^2 as a function of x: [15]

$$v^2 = v_0^2 + \frac{2}{m}\int_{x_0}^{x} F(x')\,\mathrm{d}x'.$$

Use the chain rule to prove the "$v\,\mathrm{d}v/\mathrm{d}x$ rule",

$$\dot{v} = v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{2}\frac{\mathrm{d}(v^2)}{\mathrm{d}x}.$$

Use this to rewrite Newton's second law in the separated form $m\,\mathrm{d}(v^2) = 2F(x)\,\mathrm{d}x$, and then integrate from x_0 to x. Comment on your result for the case that $F(x)$ is actually a constant.

3. A baseball is thrown vertically upward with speed v_0 and is subject to a quadratic drag of magnitude $f(v) = cv^2$. Define the terminal speed by $v_{\text{ter}} = \sqrt{mg/c}$. Write down the equation of motion for the upward journey (measuring y vertically upward) and show that it can be rewritten as [25]

$$\dot{v} = -g\left[1 + \left(\frac{v}{v_{\text{ter}}}\right)^2\right].$$

Use the "$v\,\mathrm{d}v/\mathrm{d}y$ rule" to write $\dot{v}$ as $v\,\mathrm{d}v/\mathrm{d}y$, and then solve the equation of motion by separating variables (put all terms involving v on one side and all terms involving y on the other). Integrate both sides to give y in terms of v, and hence v as a function of y. Show that the baseball's maximum height is

$$y_{\max} = \frac{v_{\text{ter}}^2}{2g}\ln\left(\frac{v_{\text{ter}}^2 + v_0^2}{v_{\text{ter}}^2}\right).$$

4. A system consists of N masses m_α at positions $\mathbf{r}_\alpha$ relative to a fixed origin O. Let $\mathbf{R}$ denote the position of the centre of mass of the system, and let $\mathbf{r}'_\alpha$ denote the position of m_α relative to the CM, that is, $\mathbf{r}'_\alpha = \mathbf{r}_\alpha - \mathbf{R}$. Assume that the internal forces between particles are equal and opposite and act along the line joining each pair of particles. [30]

   (a) Make a sketch to illustrate this last equation. [3]

   (b) Prove the useful relation $\sum_\alpha m_\alpha \mathbf{r}'_\alpha = \mathbf{0}$. Can you explain why this relation is nearly obvious? [7]

   (c) Use this relation to prove that the rate of change of the angular momentum about the CM is equal to the total external torque about the CM. (This result is surprising since the CM may be accelerating, so that it is not necessarily a fixed point in any inertial frame.) [20]

5. A particle of mass m is moving on a frictionless horizontal table and is attached to a massless string, whose other end passes through a hole in the table, where I am holding it. Initially the particle is moving in a circle of radius r_0 with angular velocity ω_0, but I now pull the string down through the hole until a length r remains between the hole and the particle. What is the particle's angular velocity now? [15]